

02LQD Specification and Simulation of Digital Systems

2020/09/14

1. Combinational circuit design (3 points)

Design a 3 input ($X_2X_1X_0$), 1 output (Z) digital logic circuit which will take all the octal digits as its input and produce the even parity bit for the corresponding octal digit.

Design in VHDL a combinational circuit implementing the above specification according to the following description styles:

- behavioural using IF THEN ELSIF statements
- behavioural using the CASE WHEN statement.

2. FSM design (12 points)

Design a garage door opener with a reset line *rst*, input lines *A*, *UP_LMT* and *DW_LMT* and output lines *MOT_UP* and *MOT_DW*. Input *A* starts the door to go up or down or stops the motion. It has thus 2 roles: it starts an action or, when an action is started, it stops it and starts the opposite one (e.g. if raising, stop raising and start lowering, or if lowering, stop lowering, start raising). Input *UP_LMT* indicates that maximum upward travel of the door has been reached. Input *DW_LMT* indicates that maximum downward travel of the door has been reached. Output *MOT_UP* causes the motor to run in direction to raise the door. Output *MOT_DW* causes the motor to run in direction to lower the door. Initially the door can be fully open or not. If not fully open, as soon as *A=1* is detected, the door raises. If fully open, as soon as *A=1* is detected, the door lowers. If fully closed, as soon as *A=1* is detected, the door raises. Whenever *A=0* is detected, motion stops and motor direction is inverted.

The FSM is a Moore machine with asynchronous reset.

- Draw the state diagram (4 points)
- Design the FSM resorting to the VHDL behavioural description style with 2 processes (4 points).
- Design the FSM resorting in Verilog with 2 always blocks (4 points).

3. FSM-D design (15 points)

Design an FSM-D to control the speed of a conveyor belt. Speed is a 4 bit output *S* that is controlled by two buttons:

- the *Up* button increases speed by 1
- the *Down* button decreases speed by 1.

If both are pushed, no change in speed occurs. Speed must initialize to 0 upon startup. Speed must remain the range 0...15. Reset is asynchronous.

Steps:

1. Draw the HLMS state diagram and write the VHDL code to capture the HLMS behavior (7 points)
2. Design a FSM-D (8 points):
 - create 2 processes for behavioural datapath description from the HLMS state diagram
 - create 2 processes for behavioural controller description from the HLMS state diagram
 - connect datapath processes to controller processes to create a single architecture.

Exam rules:

- upload on Portale della didattica in section Elaborati a .zip archive including:
 - running VHDL code and testbench
 - short report (max 3 pages) describing the solution highlighting differences w.r.t. the solution handed in at the written exam
- **no later than 2020/09/17 11.59 pm**
- format of .zip archive name surname_ID (example Rossi_244123)
- remember: no upload $\Rightarrow$ no correction $\Rightarrow$ no oral exam
- once the ranking is published on Portale della didattica in section Materiale/Results, drop an e-mail to paolo.camurati@polito.it to agree on a date for the oral exam.